APPENDIX J1
SPECIFICATIONS FOR *PROJECT MANAGER'*S SITE ACCOMMODATION
[PS Clauses 1.49 & 1.50S]

1 SCOPE

1.1 The *Contractor* shall provide and maintain the *Project Manager*'s Site Accommodation described below. All provisions for the Accommodation shall be brand new items unless otherwise specified.

1.2 The Accommodation shall be fully equipped and made ready for use within the time specified in the contract. Until such time as the Accommodation is available, the *Contractor* shall provide a temporary office for the *Supervisor* as approved by the *Project Manager* or *Supervisor*. The temporary office shall consist of minimum two (2) twenty-feet container-offices and one (1) twenty-feet container-toilets with regular cleaning and removal by specialist contractor. In addition, the facilities provided for the temporary office shall be equivalent to the requirements as stipulated in this Appendix or other provisions in the contract subject to the approval of the *Project Manager* or *Supervisor*.

1.3 The Accommodation shall comprise the following:

(a) The floor space in accordance with Clause 2.5.1 of this Appendix. The footprint of the office shall not less than 130 m^2 on plan unless otherwise approved by the *Supervisor* or *Project Manager*.

(b) Water supply, telecommunication and electricity services in accordance with the requirements of the relevant authorities.

(c) A site office sign board with the logo of the *Project Manager* and *Supervisor* and two flood lights at the entrance of the Accommodation.

(d) Not used.

(e) Covered parking within or near the premises of the Accommodation with adequate security lighting for a minimum of two vehicles or as agreed by the *Project Manager* or *Supervisor*. The cover shall be waterproofed and securely anchored for protection against strong winds/typhoons.

1.4 The Accommodation shall comprise the following:

Items	**Quantities**	**Remark**
Office-type desks with side tables, and 3-drawer movable cabinets	8	Desk and side table (with cable management): minimum dimensions 2 x 1500mm x 800mm; 3-drawer cabinet: minimum dimensions 400mm x 500mm x 650mm
Office-type chairs with arm rests	8	Seat height adjustment; back tilt adjustment; and lumbar support adjustment
4-drawer lockable file cabinets	8	Minimum dimensions 1350mm x 650mm x 450mm
4 shelves cabinets	4	Minimum dimensions 2000mm x 1000mm x 400mm
4-drawer-type safe	1	Fireproof, legal size, with lock
Conference tables	3	2m x 1m
Working tables	4	2m x 1m
Office type side chairs	12	Stackable
Utility light tables	1	800mm high, table size 800mm x 1000mm
White boards	1	2000mm x 1300mm
White boards	3	1000mm x 750mm
Cork memo board	1	2000mm x 1300mm
Outdoor notice boards	1	Weatherproof of size minimum 1200mm x 800mm
Electric binding machine.	1	Including rings and covers
Heavy duty 2-hole document punch	1	-
Heavy duty document stapler	1	-
Lamination machine up to A3 size supplied with associated consumables	1	-
USB 3.0 64GB	3	-
Electrical clocks	2	300mm diameter wall mounted
Dehumidifier	1	Capacity as approved by the *Supervisor*, with Energy Efficiency Grade 1
Ceramic Heater	2	Minimum 2000W
Refrigerator	1	400 L capacity, with Energy Efficiency Grade 1
Drinking Water filter system	2	With hot and cool water supplier and paper cups
Coffee maker	1	-
Hot water boilers	1	Wall-mounted with filter.
Microwave oven	1	Capacity as approved by the *Supervisor*
Electric kettles	1	-
Tea trolley	1	-
Glass cup, Tea cup, saucer and tea spoons	10 sets	-

Glass jug	2	-
Assorted cutlery	10	-
Electric insect killing device shall be provided	-	At least 1 no. wall-mounted at each end of the corridor
Safety/water proof lanterns	3	As approved by the *Supervisor*.
Hand torch	6	4-D cell Maglite or products having equivalent functions and performance as approved by the *Supervisor*.
5m measuring tapes	6	-
Umbrella stand	1	Capacity of 10 nos. of umbrella
Door carpet	1	3m long x 1m wide
Steel clothes rack	5	-
Plastic wastebaskets	10	500mm deep
Broom, dust pan and brush	1 set	-
First aid kit of suitable size	1	-
Mobile hanging plan racks	1	For hanging of the Drawings and working drawings in A1 drawing size
Drawing cabinets	1	For hanging of the Drawings and working drawings in A1 drawing size
Wall mounted secure key	1	For 20 keys
Cupboard	1	2000mm x 10000mm x 400mm
Hand held forehead infrared thermometer	1	-
Infrared temperature tester thermometer with floor stand	1	Sanwall – K3 Pro Infrared temperature tester thermometer or products having equivalent functions and performance as approved by the *Supervisor*.

1.5 The *Contractor* shall provide the following office equipment:-

Items	**Quantities**	**Remark**
Fax machine	1	Provide separate telephone lines for the fax machine. 1 no. facsimile line with IDD call function shall be provided with security password
Digital compact camera	1	• Minimum 9.0 mega pixels; • 28(W) – 105(T) mm zoom lens; • Auto focusing and image stabilizer; • 2.5 inch low temperature, polycrystalline silicon TFT color LCD display (Approx. 207,000 pixels); • Ancillary such as battery and charger, minimum 1G memory card.

Mobile phone	8	Smart phone with model first issued not earlier than 2021 and the accessories such as batteries, charger associated with the package provided by an authorized dealer. Each mobile phone shall be connected to an appropriate 4G service plan with at least 10GB local data per month provided by reliable service provider.
Projector for computer and screen	1	Epson EB-535W or equivalent having similar operating system, functions and performance.
Laser Presentation Remote	1	Logitech R400 Laser Presentation Remote or equivalent having similar functions and performance
Laser pointers	2	-
Video Conference Camera	1	Logitech BCC950 Conference Cam or equivalent having similar operating system, functions and performance.
Loud-speaker system	1	With 2 microphones
Shredder	1	Ideal 2601 Inchabe Gilman Office Machine or products having equivalent or similar functions and performance.
Hand calculator	8	Ten-figure, with trigonometric and memory function.

1.6 The *Contractor* shall at his own cost repair or provide a replacement within 1 week if any of the provided furniture, fixture, and equipment as stated in Clauses 1.4 and 1.5 above is malfunctioned, damaged or reported loss, save that the malfunction or damage of the photocopiers, printers, and fax machines shall be repaired or replaced within 24 hours.

1.7 In addition, the *Contractor* shall supply all the consumables from time to time as instructed by the *Supervisor* necessary for the operation and accommodation without additional cost. The *Contractor* shall replace the consumables not suitable for use.

(a) Protective clothing such as safety helmets with chin straps, safety boots, heavy raincoats with reflective strips, high visibility vests, safety rubber boots, safety belts, water proof jackets and trousers, reflective vests, ear plugs, safety harness sets, dust masks protective gloves, goggles and surgical masks;

(b) Office stationery such as business cards, stamps, automatic pencils, pens, florescent pens; adhesive tapes and stands, punchers, staplers, rulers, metric scales, set squares, hole punchers (drill type), electric pencil sharpener, paper weight, printing accessories such as ink, print cartridge, A4 and A3 size paper for photocopiers, copying and fax machines, note pads, scissors, cutters, pins, all type clips, pen holds, whiteboard pens (blue, black, red and green, all type files and folders, all type pre-inked chop, DVD-recordable discs, in/out/pending tray sets, batteries and other general stationary.

(c) Sanitary and cleaning consumables such as cleaning materials, paper towels, toilet paper, soap and hand sanitizer.

(d) Consumables for the pantry and office use, such as tea, coffee, milk and sugar, distilled water for water cooler, saucers and medical supplies to replenish first aid kit, air fresheners, disinfectants and other daily necessities.

1.8 The *Contractor* shall provide for the exclusive use of the *Project Manager* and the *Supervisor* the following services at the Accommodation:

Posts	**Quantity**
Full time amah (0830-1700 every working day with at least two years relevant working experience)	1
Full time clerical assistant (0830-1700 every working day with at least one year relevant working experience)	1

1.8.1 All provided amah and clerical assistant shall perform their respective works/duties in gear with the operations of the site accommodation and the needs of the work supervision as directed by the *Project Manager* or *Supervisor*. Due overtime allowance shall be made by the *Contractor* on the necessary overtime works of the provided amah and clerical assistant for this contract. The provided amah and clerical assistant shall be accepted by the *Project Manager* or *Supervisor*. The *Contractor* shall be responsible for substitution of all provided personnel as directed by the *Project Manager* or *Supervisor*.

1.8.2 In addition, the *Contractor* shall provide the services of Local Area Network (LAN) administration to handle the matters related to the computers and servers for the Accommodation within 24 hours after the *Supervisor* has called upon such services.

1.9 The *Contractor* shall provide pest control services once every 6 months or as other frequency necessary to keep the Accommodation habitable.

1.10 The *Contractor* shall provide 24-hour security guard services with Closed-Circuit Television (CCTV) surveillance system in the Accommodation. In addition, all external doors of the Accommodation shall be equipped with access control system approved by the *Supervisor* and under the surveillance of CCTV in the General Office.

2 DESIGN

General

2.1 The design and renovation of the Accommodation shall comply with the following:

(a) The office shall be designed and renovated according to the specified requirements and comply with the statutory requirements and requirements of the relevant authorities such as FSD, DSD, WSD, CLP, HKG etc. It shall conform to the appropriate part of the Building Ordinance. The design including but not limited to structural design, building works, electrical and mechanised works architectural works and fire services works shall be certified by an accepted by ICE.

(b) The *Contractor* shall provide an open plan office set up as far as practicable, so that the work stations shall be located abutting external windows for natural lighting and natural ventilation, whereas meeting rooms and cellular offices should be located away from external windows with glazed partitions; use of sun-shading device to block off excessive sunlight, and use of light-shelf to reflect daylight to the interior. The office plan shall be approved by the *Project Manager* or *Supervisor*;

(c) The Accommodation shall be painted/decorated internally to the satisfaction of the *Project Manager* or *Supervisor*;

(d) The Accommodation shall be connected to mains electricity, water supply, telephone and internet systems;

(e) Where main utility services are available, the *Contractor* shall make arrangements for connection and maintenance of these supplies. Where no main services is available the *Contractor* shall be wholly responsible for supplying and maintaining the utility supplies;

(f) All components and finishes shall be as specified or as reviewed by the *Project Manager* or *Supervisor* without comment;

(g) The Accommodation shall be provided with proper lightning protection and earthing system.

Building Works

2.2 Sanitary Appliances

2.2.1 Male, female and disabled toilets shall be provided at each floor with WC pans, sinks, showers, urinals (for male toilets only), and sanitary fittings. Each item of the sanitary equipment shall be individually trapped and of the self-closing type. Sanitary appliances shall be a proprietary type complete with all necessary fittings.

2.3 Pantries

2.3.1 The pantry shall be fitted with a sink with cold water supply and sewerage connection. The layout and facilities shall be subject to the *Project Manager* or *Supervisor*'s approval.

2.4 Plumbing Works

2.4.1 Pantry and toilets shall be fully plumbed for hot and cold water including the provision of electric water heaters, fixtures and fittings and all pipework ready to be connected to a water meter provided and installed by the *Contractor*. All sinks shall be fitted with hot and cold water supply. Overflow pipes for all systems and water heater shall be fitted to drain externally and independently of the waste water drainage system.

Architectural Works

2.5 Internal Partitions

2.5.1 Partitions shall be of sound, demountable construction, capable of accepting the loading of shelving, adjoining partitions and other fixtures. Unless indicated otherwise, partitions shall have a full height from floor to ceiling with acoustic insulation to a reduction of 27 dB average with a light weight door. The layout of partitions shall be shown on the Drawings submitted for review by the *Project Manager* or *Supervisor* and shall at least provide the following minimum cellular accommodation for the Accommodation:

Rooms	Min. Areas (m^2)	Quantities
Conference room	25	1
RE / Professional officer of the *Project Manager*	8.7	1
Senior Inspection Officer	6.9	1
Open plan office no. 1 - 2 Assistant Inspection Officers	8.2	1
Open plan office no. 2 - 3 Works Supervisors	6	1
General office/reception	25	1

Rooms	Min. Areas (m^2)	Quantities
Filing Room	10	1
Pantry	20	1
Toilet (Gent)	8	1
Toilet (Ladies)	8	1
Toilet (Disabled)	4	1

2.5.2 Low partition in open plan office shall be supplied and installed, including alterations during the Contract period, by the *Contractor*. The layout of the arrangement of the partitions shall be submitted to the *Project Manager* or *Supervisor* for approval together with the proposed type of partitions to be used.

2.6 Interior Finishes

2.6.1 All finishes shall be as given in the following finishes schedule, any deviation shall be approved by the *Project Manager* or *Supervisor*;

Location	**Description/Use**	**Remark**
General floor area	300mm x 300mm Vinyl floor tiles	100mm high covered vinyl skirting to walls
Ceiling	Acoustic tiles	Complete with all hanging wires, grids, wall trims and accessories.
Internal wall/partition	Emulsion paint	1 primer coat, 2 finishing coats, colour to be selected.0
Internal wall/Partition skirting	100mm high timber skirting, enamel glass	1 primer coat, 1 under coat and 2 finishing coats, colour to be selected.
External wall	Emulsion paint (exterior grade)	1 primer coat, 1 under coat and 2 finishing coats, colour to be selected.
Pantry wall	White 150mm x 150mm wall tiles	Full height
Pantry floor area	Quarry tiles	100mm high covered quarry tiles skirting, colour to be selected.
Toilet wall	White 150mm x 150mm wall tiles	Full height
Toilet floor	Granolithic	100mm high covered granolithic skirting, colour to be selected.
Internal door	Enamel paint	1 primer coat and 1 finish coat, colour to be selected. (Complete with architraves, jambs ...etc.)
External door	Enamel gloss paint (exterior grade)	1 primer coat and 2 finish coats, with jambs, architraves ...etc., colour to be selected.
External galvanised steel	Hard gloss	1 primer coat, 1 under coat and 2 finish coats, colour to be selected.

Location	Description/Use	Remark
Internal galvanised steel	Hard gloss	1 primer coat, 1 under coat and 1 finish coat, colour to be selected.
Hidden surfaces of woodwork	Polyurethane	

2.7 External Doors

2.7.1 All external doors shall be electronic lockable and shall be of exterior quality (marine ply) flush type fully weather-proof and easily cleaned with three pairs of butt hinges, spy hole, twin boltset, and 5-lever mortice lock, lever handles, laminated glass panel of suitable size on each leaf of the door and automatic overhead door closers.

2.7.2 All external doors shall be automatically locked and shall be opened by the access control system approved by the *Project Manager* or *Supervisor*.

2.8 Internal Doors

2.8.1 All internal doors shall be flushed doors of good quality having a smooth finish and easily cleaned with three pairs of butt hinges, mortice lock and knob handles, and with automatic overhead door closers to toilets and pantries only.

2.8.2 All doors shall have signs of dimensions and descriptions to be approved by the *Project Manager* or *Supervisor*.

2.9 Windows

2.9.1 Window shall be individual units. Exact size of each window unit shall be co-ordinated with the building dimensions and wherever possible, should be standardised to 750mm wide x 1200mm high for the Accommodation, with height of the windows sill at 1000mm above the finished floor level.

2.9.2 Unless otherwise specified, number of the window units to be provided shall be in accordance with the following general guidelines.

(a) No window is required for room with length of external wall less than 1m.

(b) One window shall be provided for room with length of external wall between 1m and 2m.

(c) Two windows shall be provided for room with length of external wall between 2m and 4m.

(d) Three windows shall be provided for room with length of external wall between 4m and 8m.

(e) Four windows shall be provided for room with length of external wall exceeding 8m.

(f) The following shall also be taken into account when determining the number of windows units as well as their setting out:

(i) Other window units at the approximate location on the floor above/below.

(ii) Physical constraints of the Accommodation, i.e. the structure members, building heights, etc.

(g) Window frames shall be of aluminium construction. Venetian blinds are to be fitted to all windows and glazed partitions except in the toilets.

(h) The windows are to be openable and all components, glass and sealants and the like are to be fully watertight under all conditions including wind, weather and thermal movement, with all water and condensation drained and discharged to the exterior face as necessary.

(i) For toilets and shower rooms, smaller window units at high level should be provided. Fixed wire glass louvers of 6mm thick shall be used.

(j) Window units shall be shown on the Drawings to be submitted for approval by the *Project Manager* or *Supervisor*.

(k) Fixed window guard shall be installed against burglary.

2.10 Letter Box

2.10.1 Letter box shall be fitted with external weatherproof flap and internal basket, to be of sufficient dimensions to accommodate A4 size papers and only be opened from the inside of the Accommodation.

2.11 Floor Finishes

2.11.1 In offices, pantries and corridors shall be covered in heavy duty linoleum or vinyl.

2.11.2 The floors of the toilets shall have granolithic finish laid to falls and drained at low point with a trapped gulley.

2.11.3 In toilets, the floors shall be granolithic quarry tiles under which is to be a fully waterproof membrane extended to at least the height of 60mm above the finished floor level. All finishes are laid to falls and drained at the low point with a trapped gulley. Threshold shall be provided at the doorway.

2.12 Floor Slab

2.12.1 The floor construction shall be of sound construction and with a minimum acoustic insulation reduction value of 27 dB average.

2.13 False Ceiling

2.13.1 False ceilings shall be provided to all offices, corridors, passageways, pantries, stores and where indicated on the Drawings unless otherwise approved by the *Project Manager* or *Supervisor*, these shall be installed to give a minimum headroom of 2.5m measured from the finished floor level to the soffit of the suspended ceiling tiles.

2.13.2 False ceilings shall be fully demountable acoustic 300mm x 300mm tiles, complete with all hanging wires, grids, wall trims and accessories. The design of the Accommodation shall be such that acoustic tiles shall not be dislodged by wind loading or by any differential movements caused by other means.

2.14 Toilets

2.14.1 Layouts of toilet shall be subject to approval of the *Project Manager* or *Supervisor*.

2.14.2 Wall areas surrounding wash basins and urinals shall be 20mm melamine faced marine ply with aluminium trim.

2.14.3 Proprietary shower cubicles with electrical water heater (instantaneous type 24 kW) and toilet partitions in each toilet shall be provided.

2.14.4 Samples of sanitary accessories shall be submitted to the *Project Manager* or *Supervisor* for approval prior to installation. These samples include toilet roll holders, coat hooks, paper towel holders, waste bin and mirrors of minimum size 600mm x 900mm. The exact position of these accessories shall be instructed by the *Project Manager* or *Supervisor*.

2.14.5 Wall-mounted mirrors of minimum size of 600mm x 900mm shall be provided above each wash basin.

2.14.6 Shower facilities at each floor shall be provided.

2.14.7 The following nos. of toilet cubicles, shower cubicles and wash basins shall be provided for each toilet

Facility	**Male Toilet**	**Female Toilet**
Toilet cubicles	3	2
Shower cubicles	1	0
Urinals	2	0
Wash basins	3	2

2.15 Fire Fighting Provisions

2.15.1 Adequate fire safety equipment such as sand buckets, 4.5 kg CO_2 or 9-litre water type fire extinguisher at each entrance, fire blanket and bell shall be provided in accordance with relevant ordinances and statutory requirements. Access for fire engine to be provided to reach 30m travel distance from the shed.

2.15.2 Portable fire extinguishers and all necessary accessories including wall mount steelwork and user direction display, shall be supplied and installed in positions to be approved by the *Project Manager* or *Supervisor*. 1 no. of fire extinguisher shall be provided for every $100m^2$ of floor area.

2.15.3 All portable fire extinguishers shall be approved type certified by Fire Services Department and regularly maintained and inspected by the *Contractor*.

2.15.4 Simple fire alarm comprising break glasses, smoke detector (1 no. detector head per $200m^2$ of floor area) and bells shall be installed. Design of the system shall be proposed by the *Contractor* and approved by the *Project Manager* or *Supervisor*.

E&M Installations

2.16 General

2.16.1 The Accommodation shall be wired for electrical light, power, telephone and computer data points for the Local Area Network (LAN). The complete electrical works/installation shall comply with the current supply rules of the respective supply company.

2.17 Air Conditioning

2.17.1 Fan coil air conditioning with reverse cycle shall be provided, capable of maintaining a maximum temperature of not higher than 25.5℃ at all locations in the Accommodation during high summer. The air conditioning system shall be subdivided into sufficient local zones, each complete with individual local temperature controls. In addition, the pantries and toilets shall have appropriately sized and mounted extract fans. All the air-conditioners provided shall have Energy Efficiency Grade 1. The *Contractor* shall provide household electric fans in addition to air-conditioners to improve the air circulation where necessary and achieve even cooling throughout the intended area efficiently.

The electric fans shall be with fan blades of sweep diameter ranging from 305 to 415 mm with low, medium and high speed, and timer control.

2.18 Fluorescent Lighting

2.18.1 The *Contractor* shall adopt lighting zone control. The minimum number of lighting control points for any space that is classified as an office should comply with the requirements in Table (LG5) of Building Energy Code (Lighting Installations) issued by EMSD. Stairways and passages shall be properly lit and equipped with automatic emergency lights.

2.18.2 For each individual functional area or functional activity in a multi-functional space, separate lighting control points should be provided to allow operation of the luminaires for that area or activity only.

2.18.3 High energy efficiency lamps for general lighting unless otherwise approved by the *Project Manager* or *Supervisor*. The lighting power density of a space of a type classified in Table 5.4 of Building Energy Code (Lighting Installation) should not exceed the corresponding maximum allowable value, unless the total electrical power consumed by the complete fixed lighting installations in the space does not exceed 100W.

2.18.4 The portable table lamp shall comply with "IEC 60598-1: Luminaires, Part 1 – General Requirements and Tests" and "IEC 60598-2-4: Part 2 Particular Requirements –Section 4: Portable General Purpose Luminaires: whenever appropriate.

2.19 Power Outlet

2.19.1 Each room shall be equipped with adequate 13 amp power outlets at positions agreed with the *Project Manager* or *Supervisor*, but at a rate of not less than 1 no. double power socket per 4 square metres of floor area.

2.20 Telephone

2.20.1 Telephone wiring shall be installed in conduits from a central trunking serving the telephone inlet and shall be in accordance with requirements of the telecommunication company.

2.20.2 Telephones with speaker phone and conferencing facilities shall be provided to each work station and the telephone system shall include voice mail and direct dial-in facility for each telephone. 8 no. incoming telephone lines, all with IDD call facility barred, shall be provided. 1 no. facsimile line with IDD call function shall be provided with security password.

2.21 Energy Saving

2.21.1 The electrical appliances proposed for approval by the *Project Manager* or *Supervisor* shall be those with Energy Efficiency Grade 1, under the Energy Efficiency Labelling Scheme. Other suitable energy efficient process should also be considered.

2.21.2 The *Contractor* shall incorporate metering devices for measuring current and total energy consumption for any incoming power supply, single or three phases.

2.21.3 The *Contractor* shall incorporate occupancy sensors to the lighting installation in open plan office, cellular office, store rooms and common areas to provide automatic control and shut off lights when office accommodation is not occupied or in use.

2.21.4 The *Contractor* shall incorporate timer switches for control and automatic shut off of all non-essential power sockets, lighting, water heaters, water circulation pumps and air-conditioners.

2.22 Water Saving

2.22.1 The water consuming appliances proposed for approval by the *Project Manager* or *Supervisor* shall be those with Water Efficiency Grade 1, under the Water Efficiency Labelling Scheme. Other suitable water efficient process should also be considered.

2.22.2 The Contractor shall use dual flush cisterns for water closets.

Particular Specification for Sustainable Designs and Features in Resident Site Staff's Site Accommodation

General Requirements

1. This site accommodation shall be of low-cost design.

2. Used materials and products which are in reasonable and serviceable conditions should be allowed. If used materials and products are not available, the materials and products shall be supplied from regional manufacturers to minimize transportation cost and carbon footprint.

3. The *Contractor* shall design , supply, install, service and maintain all site accommodation together with the sustainable provisions in accordance with the requirements as stated in this Particular Specification (PS) and the latest edition of Code of Practice for Energy Efficiency of Building Services Installation (hereafter referred as Building Energy Code) issued by EMSD.

Sustainable Provisions

The *Contractor* shall comply with the following requirements on design, installation and management for the site accommodation:

A. Design

(i) The *Contractor* shall provide an open plan office set up as far as practicable, so that the work stations shall be located abutting external windows for natural lighting and natural ventilation, whereas meeting rooms and cellular offices should be located away from external windows with glazed partitions; use of sun-shading device to block off excessive sunlight, and use of light-shelf to reflect daylight to the interior.

(ii) The *Contractor* shall plan and set the site accommodation in accordance with the study of the microclimate, which includes the consideration on the orientation of the office to enhance cross ventilation and maximize the use of natural light; and the position of the office to take advantage of natural shelter from excessive daylight and nuisance (e.g. noise and dust).

(iii) The *Contractor* shall provide double glazed windows with solar film and/or exterior shading devices at the east and west elevations.

B. Energy Saving

(i) The *Contractor* shall adopt lighting zone control. The minimum number of lighting control points for any space that is classified as an office should comply with the requirements in Table 5.5 under the latest edition of "Code of Practice for Energy Efficiency of Building Services Installation" (i.e. Building Energy Code) issued by EMSD. For each individual functional area or functional activity in a multi-functional space, separate lighting control points should be provided to allow operation of the luminaires for that area or activity only.

(ii) The *Contractor* shall adopt light-emitting diode (LED) type high energy efficiency lamps for general lighting unless otherwise accepted by the Project Manager. The lighting power density of a space of a type classified in Table 5.4 under the latest edition of Building Energy Code should not exceed the corresponding maximum allowable value, unless the total electrical power consumed by the complete fixed lighting installations in the space does not exceed 70W.

(iii) Unless otherwise accepted by *Project Manager* or *Supervisor*, the *Contractor* shall supply energy efficient equipment with energy efficiency labels of Grade 1 or Grade 2 for "grading type" or of "recognition type" if the Equipment falls within the category of either Mandatory or Voluntary Energy Efficiency Labelling Scheme under EMSD.

(iv) The *Contractor* shall incorporate metering devices for measuring current and total energy consumption for any incoming power supply, single or three phases.

(v) The *Contractor* shall incorporate occupancy sensors to the lighting installation in open plan office, cellular office, store rooms and common areas to provide automatic control and shut off lights when office accommodation is not occupied or in use.

(vi) The *Contractor* shall incorporate timer switches for control and automatic shut off of all non-essential power sockets, lighting, water heaters, water circulation pumps and air-conditioners.

(vii) Air-conditioners should be set to run at a temperature of not less than 25.5^{o}C for energy saving. The *Contractor* shall provide household electric fans in addition to air-conditioners to improve the air circulation where necessary and achieve even cooling throughout the intended area efficiently. The electric fans shall be with fan blades of sweep diameter ranging from 305 to 415mm with low, medium and high speed, and timer control.

(viii) The *Contractor* shall incorporate task lighting design concept in open plan office and cellular office without meeting facilities. A lower ambient lighting level of 300 lux average shall be provided and each task plane shall be supplemented with a portable table lamp to achieve 500 lux average.

(ix) The portable table lamp shall comply with "IEC 60598-1: Luminaires, Part 1 – General Requirements and Tests" and "IEC 60598-2-4: Part 2 Particular Requirements – Section 4: Portable General Purpose Luminaires" whenever appropriate.

C. Materials

(i) The *Contractor* shall use standard composited wall panels and other reusable materials, such as GMS steel frame and insulated panel as far as practicable.

(ii) Ozone Depleting Substances and materials shall not be used.

(iii) The *Contractor* shall provide office furniture / equipment arrangements in modular dimensions so that building material, furniture and equipment can be reused in the future; easy fixing by bolt-and-nut facilitates future demounting and reusing of building components to reduce construction waste.

(iv) The *Contractor* shall use low embodied energy materials such as paving blocks made from recycled aggregate, glass and fly ash.

(v) The *Contractor* shall provide printing and photocopying papers accredited with internationally recognized green forestry certificates/labels, including the Forest Stewardship Council (FSC) or Programme for the Endorsement of Forestry Certification (PEFC), as set out in Environmental Bureau Circular Memorandum No. 6/2015.

D. Water Saving

(i) The *Contractor* shall use water saving products registered under the "Water Efficiency Labelling Scheme" (WELS) of Water Supplies Department (WSD) where feasible and applicable.

(ii) The *Contractor* shall use water closets with dual flush function.

(iii) The *Contractor* shall provide automatic sensing flushing system for urinal bowls.

(iv) For those sites with technical constraint in obtaining flushing water connection, waterfree urinals can be adopted and the *Contractor* shall arrange cleaning, management and replacement of the cartridge or scaling liquid or the like all in accordance with manufacturer's instruction.

(v) The *Contractor* shall adopt rainwater harvesting and recycling system for toilet flushing / planting watering / other site uses to reduce water consumption. The scope and scale of works with full justifications and all design works including sizing, calculation, technical information, data, schematic and detailed layout shall be submitted to Project Manager for approval. Reference shall be made to the "Technical Specifications on Grey Water Reuse and Rainwater Harvesting" issued by the Water Supplies Department for the design, installation, commissioning, operation and maintenance of grey water reuse and rainwater harvesting systems.

E. Waste Reduction

(i) Selling of plastic bottled water (one litre or less) in all vending machines within the construction site shall not be allowed. At the same time, it is also encouraged to minimize the use of plastic bottled water as far as possible.

F. Housekeeping

(i) The *Contractor* shall maintain the site accommodation together with all the sustainable provisions in serviceable conditions.

(ii) The *Contractor* shall ensure that facilities for natural lighting and ventilation are not adversely affected or obstructed.

(iii) The *Contractor* shall prepare that the automatic devices for energy saving are not overridden manually or by other means.

(iv) The *Contractor* shall prepare an inspection checklist for checklist for checking and monitoring the conditions and functions of the facilities in the site accommodation.

(v) The *Contractor* shall assign a staff member at supervisory level or above to conduct the checking and arrange the corrective actions as and when necessary.

(vi) The *Contractor* shall post notices at conspicuous locations to remind staff to switch off the computers and other electrical appliances when not in use and before leaving.

(vii) The *Contractor* shall prepare and submit monthly summary reports which should include details of the checking, the findings and the corrective actions taken, if any, in the reporting month to the *Project Manager* or *Supervisor* for record and monitoring.

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